

# NAEOS CLI Reference

Version 3.5.0

NAEOS Foundation

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NAEOS provides a complete CLI with commands for every stage of the engineering pipeline.

## Core Commands

| Command                       | Description                                       |
|-------------------------------|---------------------------------------------------|
| <code>naeos run</code>        | Execute the full pipeline                         |
| <code>naeos validate</code>   | Validate a specification using the NAEOS pipeline |
| <code>naeos ai compile</code> | Compile a spec into AI instruction sets           |
| <code>naeos context</code>    | Generate AI context bundles from specifications   |
| <code>naeos test</code>       | Run tests for generated code                      |
| <code>naeos docgen</code>     | Generate documentation from specification         |

## Project & Spec Commands

| Command                        | Description                                                       |
|--------------------------------|-------------------------------------------------------------------|
| <code>naeos init</code>        | Initialize a new NAEOS project or generate config                 |
| <code>naeos create</code>      | Interactive project creation wizard                               |
| <code>naeos scaffold</code>    | Generate a starter project scaffold                               |
| <code>naeos import</code>      | Import specifications from HCL format to NAEOS<br>YAML/JSON       |
| <code>naeos export</code>      | Export generated artifacts to a directory                         |
| <code>naeos audit</code>       | Security audit of generated or source files                       |
| <code>naeos diff</code>        | Compare generated artifacts with existing output directory        |
| <code>naeos repair</code>      | Repair the NAEOS output directory                                 |
| <code>naeos lint</code>        | Lint a specification file                                         |
| <code>naeos inspect</code>     | Inspect the NAEOS pipeline result                                 |
| <code>naeos preview</code>     | Preview generated artifacts without writing them                  |
| <code>naeos distributed</code> | Run pipeline tasks in distributed mode across multiple<br>workers |

## Build, Deploy & CI/CD

| Command                  | Description                           |
|--------------------------|---------------------------------------|
| <code>naeos build</code> | Build artifacts from a specification  |
| <code>naeos cicd</code>  | Generate CI/CD pipeline configuration |
| <code>naeos cloud</code> | Cloud deployment commands             |

| Command                    | Description                                                 |
|----------------------------|-------------------------------------------------------------|
| <code>naeos deploy</code>  | Deploy the pipeline output to a target environment          |
| <code>naeos gateway</code> | API gateway management                                      |
| <code>naeos graphql</code> | Start GraphQL API server                                    |
| <code>naeos helm</code>    | Helm chart scaffolding and validation                       |
| <code>naeos airgap</code>  | Create, inspect, and import air-gapped distribution bundles |

## Management Commands

| Command                        | Description                                                           |
|--------------------------------|-----------------------------------------------------------------------|
| <code>naeos mcp</code>         | Start MCP (Model Context Protocol) server                             |
| <code>naeos marketplace</code> | Browse and install templates, profiles, and plugins                   |
| <code>naeos profile</code>     | Manage industry-specific project profiles                             |
| <code>naeos plugin</code>      | Manage NAEOS plugins                                                  |
| <code>naeos template</code>    | Manage generation templates, prompt library, and template marketplace |
| <code>naeos workspace</code>   | Manage multi-module workspaces                                        |
| <code>naeos artifacts</code>   | Manage generated project artifacts                                    |
| <code>naeos migrate</code>     | Manage spec schema migrations                                         |
| <code>naeos migration</code>   | Database migration management                                         |
| <code>naeos rollback</code>    | Rollback to a previous snapshot of generated artifacts                |
| <code>naeos lock</code>        | Manage lock files for reproducible builds                             |

## Monitoring & Operations

| Command                          | Description                                                 |
|----------------------------------|-------------------------------------------------------------|
| <code>naeos status</code>        | Show current pipeline, system and project status            |
| <code>naeos doctor</code>        | Run diagnostics on the NAEOS environment and configuration  |
| <code>naeos watch</code>         | Watch for specification changes and re-run the pipeline     |
| <code>naeos kernel</code>        | Inspect the NAEOS kernel and service registry               |
| <code>naeos dashboard</code>     | Start NAEOS web dashboard                                   |
| <code>naeos api</code>           | Start NAEOS REST API server                                 |
| <code>naeos health</code>        | Run system health checks and diagnostics                    |
| <code>naeos monitor</code>       | Start monitoring server with Prometheus metrics             |
| <code>naeos observability</code> | Observability and telemetry management                      |
| <code>naeos events</code>        | Event sourcing commands for pipeline audit trail and replay |
| <code>naeos history</code>       | Show pipeline run history from persisted events             |
| <code>naeos serve</code>         | Run NAEOS as a production daemon (HTTP/TLS, systemd)        |

## Security & Compliance

| Command                       | Description                                                         |
|-------------------------------|---------------------------------------------------------------------|
| <code>naeos auth</code>       | Authentication and authorization management                         |
| <code>naeos security</code>   | Security and secrets management                                     |
| <code>naeos compliance</code> | Compliance reporting and audit log export                           |
| <code>naeos config</code>     | Configuration management and resolution (env/file/K8s secret/Vault) |

| Command                     | Description                                            |
|-----------------------------|--------------------------------------------------------|
| <code>naeos policy</code>   | Manage governance policies                             |
| <code>naeos control</code>  | Governance control plane evaluation                    |
| <code>naeos evidence</code> | Immutable evidence store — governance audit trail      |
| <code>naeos verify</code>   | Independently verify governance evidence               |
| <code>naeos runtime</code>  | Runtime gateway for authorized agent execution         |
| <code>naeos sign</code>     | Sign and verify artifacts (Ed25519)                    |
| <code>naeos sbom</code>     | Software Bill of Materials generation and verification |

## Data & Integration

| Command                     | Description                                  |
|-----------------------------|----------------------------------------------|
| <code>naeos db</code>       | Database connection and migration management |
| <code>naeos broker</code>   | Message broker management                    |
| <code>naeos ws</code>       | Start WebSocket server for real-time updates |
| <code>naeos lsp</code>      | Start NEIR Language Server Protocol server   |
| <code>naeos supabase</code> | Supabase backend management                  |

## Developer Experience & Utility

| Command                       | Description                        |
|-------------------------------|------------------------------------|
| <code>naeos ai</code>         | AI-powered assistance commands     |
| <code>naeos dx</code>         | Developer experience tools         |
| <code>naeos tui</code>        | Terminal user interface tools      |
| <code>naeos schema</code>     | NEIR schema registry operations    |
| <code>naeos search</code>     | Full-text search engine management |
| <code>naeos perf</code>       | Performance optimization tools     |
| <code>naeos benchmark</code>  | Run pipeline benchmarks            |
| <code>naeos docs</code>       | Generate project documentation     |
| <code>naeos completion</code> | Generate shell completion scripts  |
| <code>naeos version</code>    | Show NAEOS version                 |

For detailed usage of each command, run `naeos <command> --help`.

## Download

- CLI Reference PDF